Billy Joel Catalog Analytics

Popularity, career trends and forecasting across Billy Joel's 12 studio albums (1971-1993)

Project 4 - Analyze Data with Power BI | Eylon Lavi

Analysis Topics

Catalog overview | Career timeline | Album performance | Song analytics | Trend forecasting

Research Questions

1. How did song popularity evolve across the career - which eras and albums peaked?
2. Which albums generate the hits, and how concentrated are they?
3. Does song length have any relationship with popularity?
4. Is there a measurable long-term trend, and what would it predict for a hypothetical next album?
5. Which songs are unusually long outliers, and does their length hurt their popularity?

About the Data & Model

Source: Songs.csv - 115 rows x 5 columns (title, album, release year, popularity 0-100, duration in ms).
Cleaning in Power Query: CSV quote handling, trimming, type enforcement, range validation, imputation rules, removal of the non-album single and a duplicate (115 -> 113 songs), engineered tiers, bands and IQR outlier flags.
Model: star schema - FactSongs + DimAlbum + DimYear, What-If 'Hit Threshold', regression table and 60 DAX measures in display folders; 3 hierarchies.

How to Use

Navigation bar on every page | Slicers are synced across pages | Right-click an album > Drill through > Album Detail | Hover album columns for a tooltip page | Bookmark buttons switch views | Ctrl+click buttons in Desktop

113	12	52.0	17
Unique songs	Studio albums	Average popularity (0-100)	Hit songs (popularity >= 70)

Executive Overview | **Career Timeline** | **Album Performance** | **Song Analysis** | **Trend & Forecast** | **Key Findings**

Data source: Songs.csv (Billy Joel discography with Spotify-style popularity scores). Popularity is a relative 0-100 score, not sales or streams. Built with Power Query, DAX (incl. LINESTX regression) and PBIR report definition.

Executive Overview

Catalog size, popularity and hit performance at a glance

Career Era: All

Decade: All

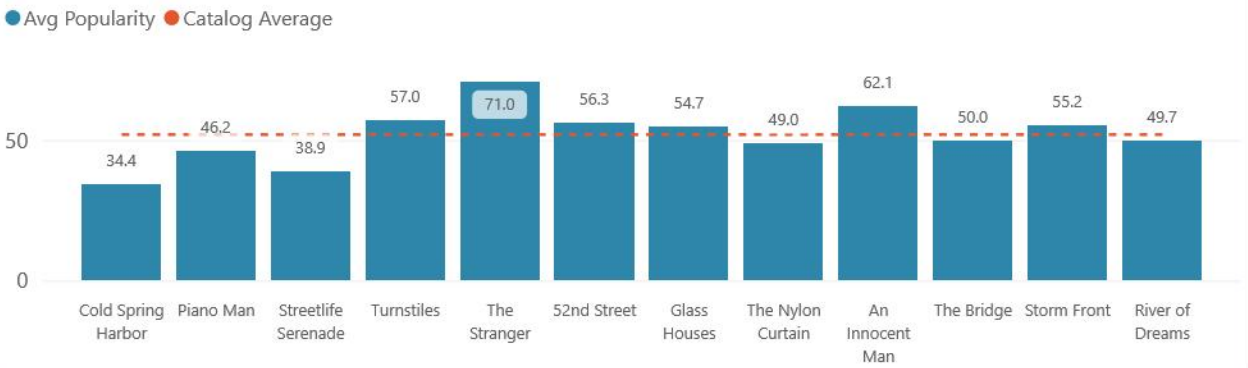
Hit threshold (score ≥, default 70): All

Showing: All eras | All decades

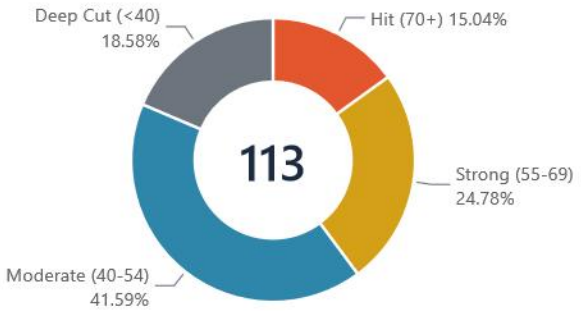
Current filter context

113	12	52.0	17	15.0%	8.0
Songs	Studio Albums	Avg Popularity	Hit Songs	Hit Rate	Runtime (hours)

Average Popularity by Album vs Catalog Average (hover for album tooltip)



Songs by Popularity Tier



Latest Album vs Catalog Average

49.7!

Goal: 52.0 (-4.34%)

Hit Rate vs 25% Target



Key Insights (full catalog)

- 113 songs across 12 studio albums; average popularity 52.0 (median 48).
- The Stranger (1977) is the standout album: avg popularity 71 and 6 of the 17 hits (35%).
- Only 15% of songs reach hit level (70+); 42% are moderate (40-54) - a classic long-tail catalog.
- Raise the Hit Threshold slicer to stress-test how concentrated the hits are.

Career Timeline

Release-over-release, yearly and decade trends

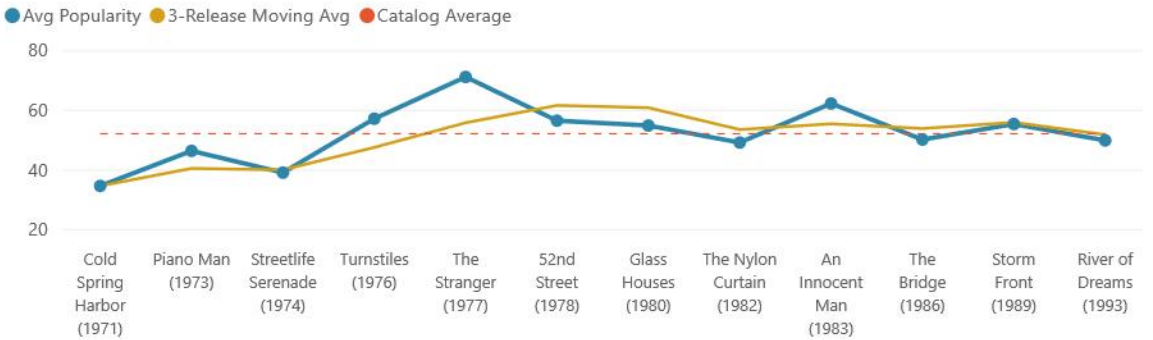
Decade: All

Career Era: All

Insight

Peak era 1977-83 averages 58.6 vs 43.7 in the Early Years; later albums oscillate around the mean (52.0).

Popularity Trajectory by Release (with 3-release moving average)



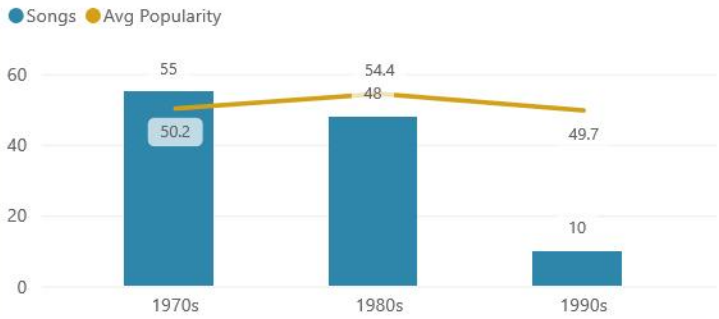
Decade > 5-Year Period > Year (expand with +)

Decade	Songs	Avg Pop.	% Songs	Hits	vs Prev.
⊞ 1970s	55	50.2	48.7%	9	
⊞ 1980s	48	54.4	42.5%	7	
⊞ 1990s	10	49.7	8.8%	1	-5.5
Total	**113**	**52.0**	**100.0%**	**17**	

Release-over-Release Change in Popularity



Songs & Popularity by Decade (drill down to Year)



Album Performance

Ranking, hit concentration and drill-through to album detail

Career Era: All

Hit threshold (score ≥, default 70): All

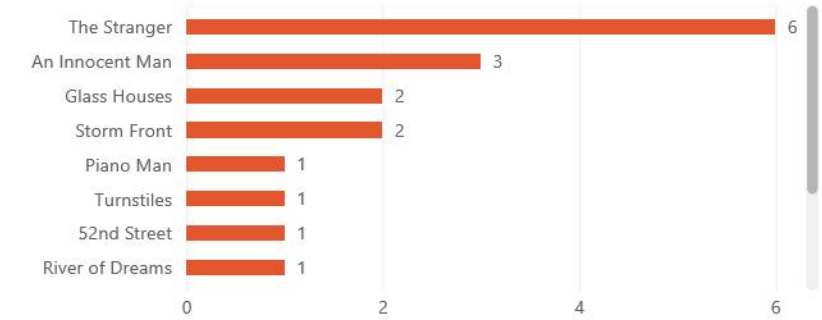
Hit Songs view | Bubble view

Tip
Right-click any album > Drill through > Album Detail

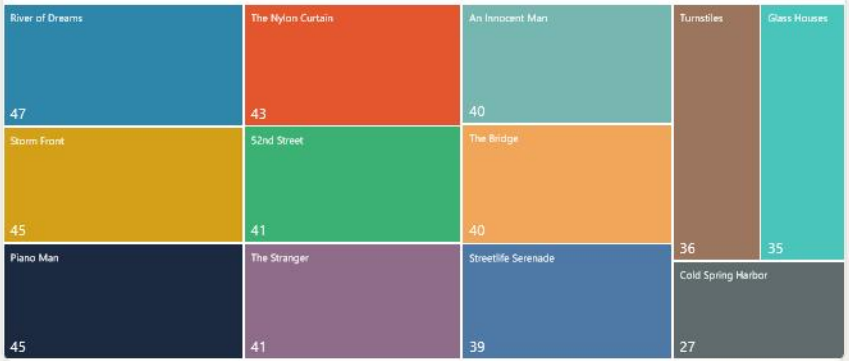
Album Scorecard - Era > Album (expand with +)

Era	Songs	Avg Pop.	Rank	Hits	Hit Rate	% of All Hits	Top Song
⊞ Early Years (1970-1976)	37	43.7	2	5.4%	11.8%	Piano Man
⊞ Commercial Peak (1977-1983)	47	58.6	12	25.5%	70.6%	Vienna
⊞ Late Studio Era (1984-1993)	29	51.7	3	10.3%	17.6%	We Didn't Start the Fire
Total	**113**	**52.0**	**17**	**15.0%**	**100.0%**	**Vienna**

Hit Songs (popularity >= 70)



Total Runtime by Album (minutes)



Song Analysis

Does song length drive popularity? Correlation, regression and outliers

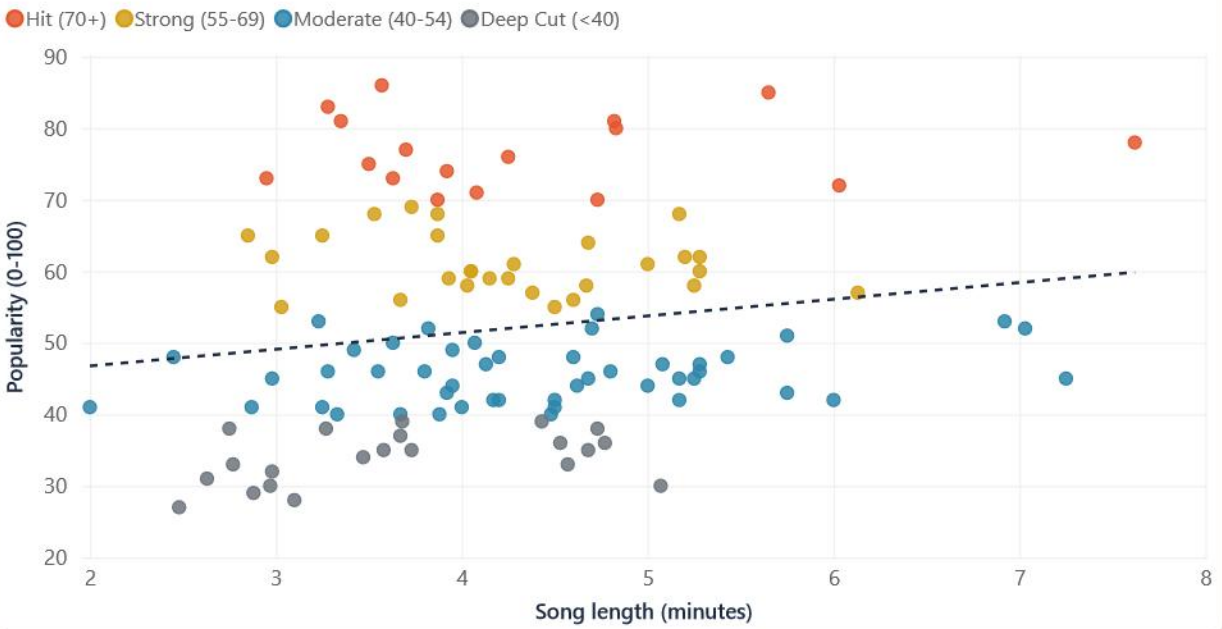
0.17	Weak positive relationship	+2.3	4:14	4	57.0
Pearson r (duration vs popula...	Interpretation	Points per extra minute	Average song length	Duration outliers (IQR rule)	Avg popularity of outliers

Career Era: All

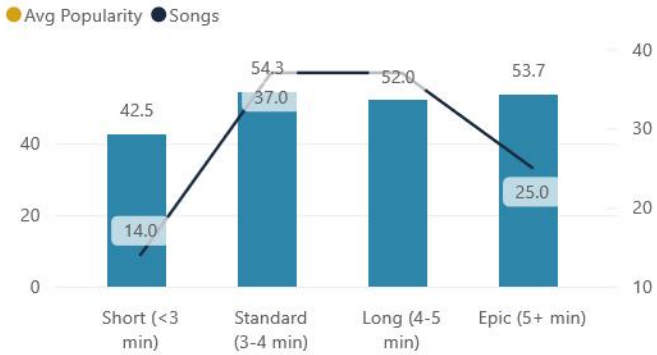
Popularity Tier: All

Scatter view | Song table view

Every Song: Length vs Popularity (dashed = linear trend line)



Popularity by Song-Length Band



Findings (full catalog)

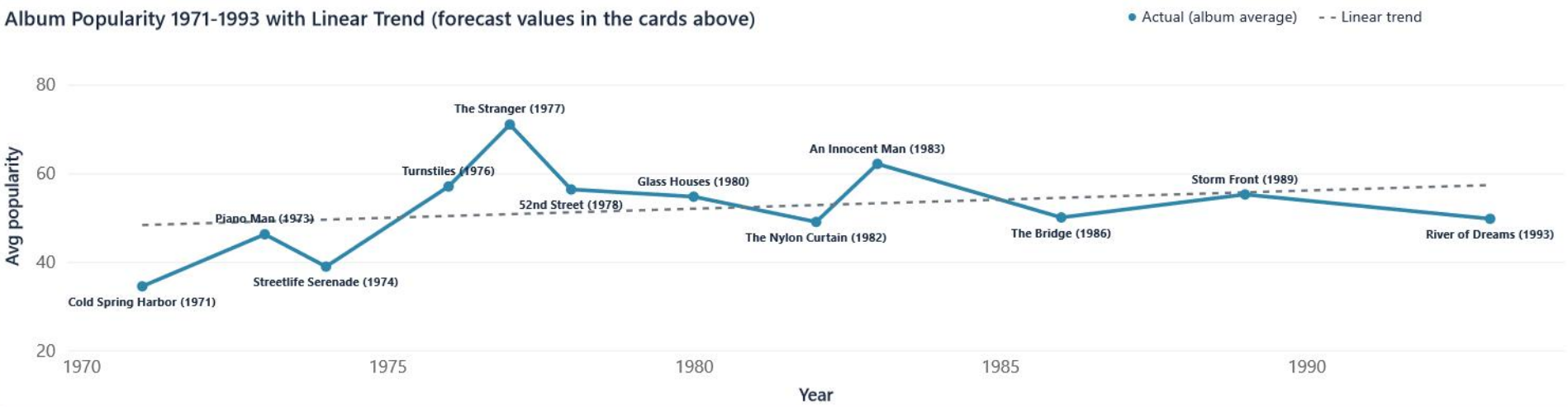
- Weak positive link: r = 0.17; LINESTX slope = +2.3 popularity points per extra minute.
- Short songs (<3 min) are the least popular group (42.5 vs 52-54 for longer bands).
- 4 IQR outliers, all over 6:50 - incl. 'Scenes from an Italian Restaurant' (7:37, popularity 78).
- Length alone does not make a hit: hits appear in every band.

Trend & Forecast

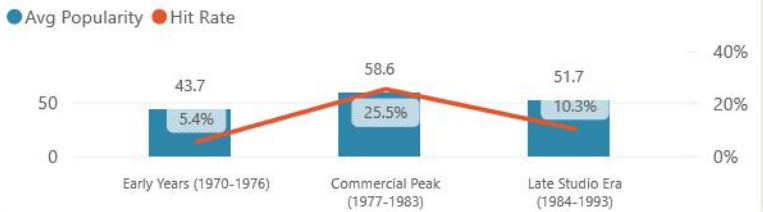
DAX LINESTX regression with a 95% prediction interval

+4.1	7.9%	9.86	1995	58.1	31.0 - 85.3
Trend: points per decade	R-squared (fit quality)	Std. error (points)	Hypothetical next album	Forecast popularity	95% prediction range

Album Popularity 1971-1993 with Linear Trend (forecast values in the cards above)



Pattern by Era: Popularity & Hit Rate



Linear trend over 12 studio albums: +4.1 popularity points per decade, R^2 = 8%. Next album (1995) forecast: 58.1 (95% range 31.0 - 85.3).

Model summary - calculated live in DAX

Method & Caveat

OLS regression of album-average popularity on release year (12 studio albums, DAX LINESTX).
95% interval = t(0.975, n-2) x SE x sqrt(1 + 1/n + (x - mean)^2 / Sxx).
R-squared ~8%: time explains little of popularity - treat the forecast as directional only.

Album Detail: Streetlife Serenade (1974)

Click an album in the list to change

Back to report

10	38.9	11	-13.1	0	The Entertainer
Songs	Avg Popularity	Rank in catalog (of 12)	vs catalog average	Hit Songs	Most popular song

Pick an album

Album

- All albums
- Cold Spring Harbor
- Piano Man
- Streetlife Serenade
- Turnstiles
- The Stranger
- 52nd Street
- Glass Houses
- The Nylon Curtain
- An Innocent Man
- The Bridge
- Storm Front
- River of Dreams

Song List

#	Song	Length	Popularity	Tier	Length Band
1	Streetlife Serenader	5:17	46	Moderate (40-54)	Epic (5+ min)
2	Los Angelenos	3:41	39	Deep Cut (<40)	Standard (3-4 min)
3	Great Suburban Showdown	3:44	35	Deep Cut (<40)	Standard (3-4 min)
4	Root Beer Rag	2:59	45	Moderate (40-54)	Short (<3 min)
5	Roberta	4:32	36	Deep Cut (<40)	Long (4-5 min)
6	The Entertainer	3:38	50	Moderate (40-54)	Standard (3-4 min)
7	Last of the Big Time Spenders	4:34	33	Deep Cut (<40)	Long (4-5 min)
8	Weekend Song	3:28	34	Deep Cut (<40)	Standard (3-4 min)
9	Souvenir	2:00	41	Moderate (40-54)	Short (<3 min)
10	The Great Tower	5:04	30	Deep Cut (<40)	Epic (5+ min)
Summary		**Total length: 38:57**	**Avg popularity: 38.9**	**Album tier: Deep Cut (<40)**	**Length band: Standard (3-4 min)**

Songs by Popularity



Key Findings Summary

Answers to the five research questions - full catalog

113	12	52.0	15.0%	0.17	58.1
Songs analysed	Studio albums (1971-1993)	Average popularity	Hit rate (popularity >= 70)	Length vs popularity (r)	Next album forecast (1995)

1. Career evolution - when did popularity peak?

- Popularity peaked in the Commercial Peak era (1977-1983): average 58.6, vs 43.7 in the Early Years and 51.7 in the Late Studio Era.
- The Stranger (1977) is the high point at 71.0, followed by An Innocent Man (1983) at 62.1.
- The early albums are the weakest - Cold Spring Harbor (1971) has the lowest score, 34.4.
- From 1984 on, albums stay close to the catalog mean (52.0).

Pages: Overview, Timeline

2. Where do the hits come from?

- 17 of 113 songs (15.0%) reach hit level (70+) - below the 25% target.
- Hits are highly concentrated: the 1977-1983 era holds 12 of the 17 (70.6%), and The Stranger alone has 6 (35.3%).
- 4 of the 12 albums have no hit at all: Cold Spring Harbor, Streetlife Serenade, The Nylon Curtain and The Bridge.
- A long-tail catalog: 42% of songs are Moderate (40-54) and 19% are Deep Cuts (<40).

Pages: Overview, Albums

3. Does song length matter?

- Only a weak positive link: Pearson r = 0.17.
- LINESTX regression slope: about +2.3 popularity points per extra minute.
- Short songs (<3 min) are the least popular group: 42.5, vs 52-54 for every longer band.
- Hits appear in every length band - length alone does not make a hit.

Page: Songs

4. Long-term trend & forecast

- Linear trend across the 12 studio albums: +4.1 popularity points per decade.
- R-squared is only 7.9% - the release year explains very little of an album's popularity.
- A hypothetical next album (1995) is forecast at 58.1, but the 95% prediction range is 31.0 - 85.3.
- The forecast is directional only: which album it is matters more than when it was released.

Page: Forecast

5. Long-song outliers

- 4 songs are duration outliers by the IQR rule, all longer than 6:50: Scenes from an Italian Restaurant, Captain Jack, Goodnight Saigon and Until the Night.
- Long songs are not penalised: the outliers average 57.0, above the catalog mean of 52.0.
- 'Scenes from an Italian Restaurant' (7:37) scores 78 - a hit despite its length.
- Each outlier scores within 7 points of its own album's average - the album, not the length, sets the level.

Page: Songs

Bottom line

- Billy Joel's catalog popularity is driven by one era: 1977-1983 has the highest average (58.6) and 70% of the hits.
- Neither release year nor song length explains much - the specific album and song matter most.
- A small core of 17 hits carries the catalog; 85% of the songs are below hit level.
- Recommendation: build playlists and promotion around The Stranger and the Commercial Peak albums.

All values: full catalog, no filters, hit threshold 70